

REC • CAM 07 • AISLE 07

COMPUTER VISION • GENERATIVE-MODELS PROJECT

SentinelFrame

Frame-Level Hand-State Recognition for Retail CCTV



What it is — and deliberately isn't



A per-frame perception layer

- ✓ Reads what the hands are doing
- ✓ Each label is scored on its own
- ✓ Built on a pretrained vision model
- ✓ Built to feed downstream systems



Not a theft detector

- ✗ It won't tell you who's shoplifting
- ✗ It has no memory of past frames
- ✗ It doesn't track or identify anyone
- ✗ The judgment call happens downstream

Multi-label classification, one frame at a time



8

observable
hand states



σ

sigmoid per label
(independent)



BCE

with pos-weighted
logits loss

EXAMPLE OUTPUT — one frame

hand_in_pocket	0.91
object_in_hand	0.73
interacting_with_shelf	0.68
hand_occluded_generic	0.05
hand_in_bag	0.04
hand_under_clothing	0.01
both_hands_not_visible	0.00

Three states co-active — exactly what multi-label is for.

**No temporal information ever
enters the model.**

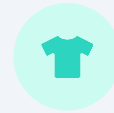
Eight observable states



hand_in_pocket



hand_in_bag



hand_under_clothing



object_in_hand



interacting_with_shelf



hand_occluded_generic



both_hands_not_visible



no_visible_hand_interaction

Every training image is generated



hand_in_pocket



hand_in_bag



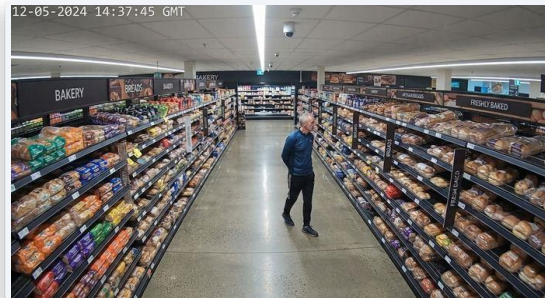
hand_under_clothing



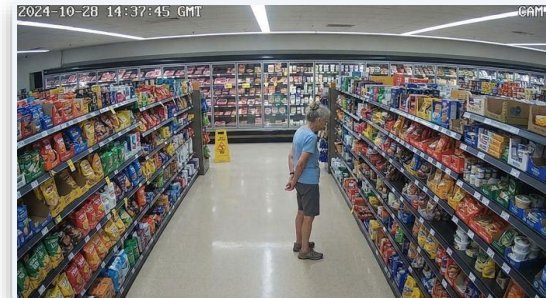
object_in_hand



interacting_with_shelf



hand_occluded_generic



both_hands_not_visible



no_visible_hand_interaction

Prompts assembled from randomized parts

base_context

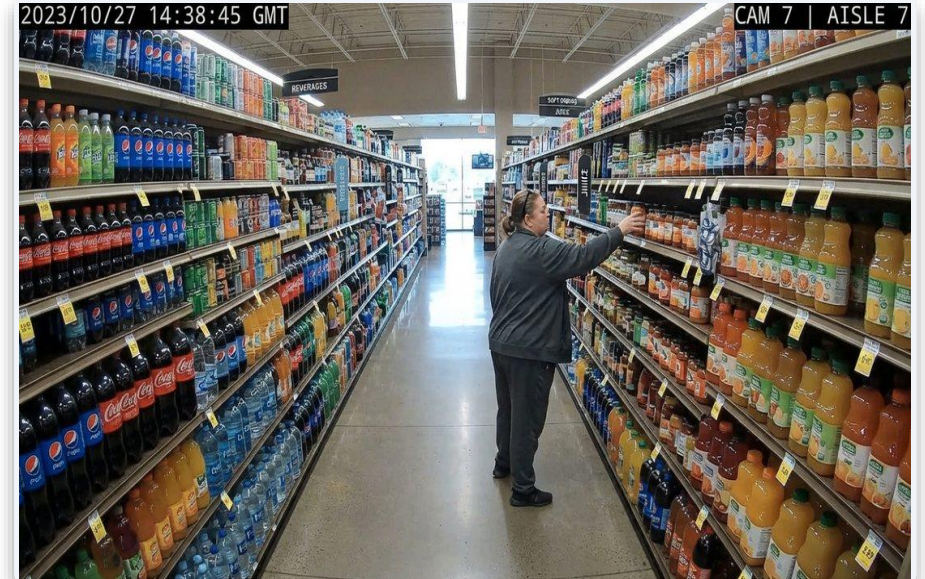
"...captured by {camera_view} ... {lighting} ... a solitary {age} {subject} wearing {clothing_style} ... {camera_imperfections}"

action

"...{hand_side} hand naturally resting inside the {pocket_location}..."

framing

"Full-body CCTV-style surveillance view from above..."



Generated sample – interacting_with_shelf

Gemini image model

structured prompt → one labelled 16:9
CCTV frame

Built like a production job, not a script



Reproducible from a seed



Resumes after a stop



Retries on API errors



Parallel API workers

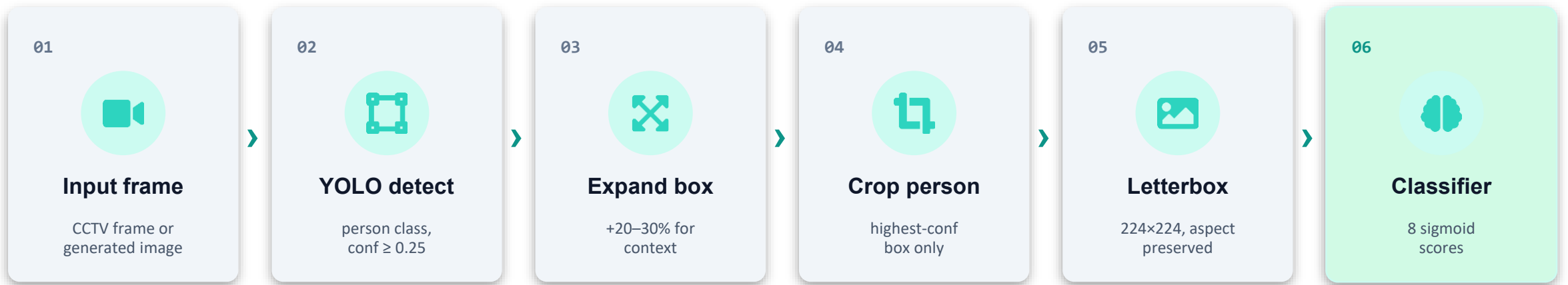


Metadata for every image



Balanced across classes

From raw frame to a 224×224 crop



From raw frame to a 224×224 crop: an example

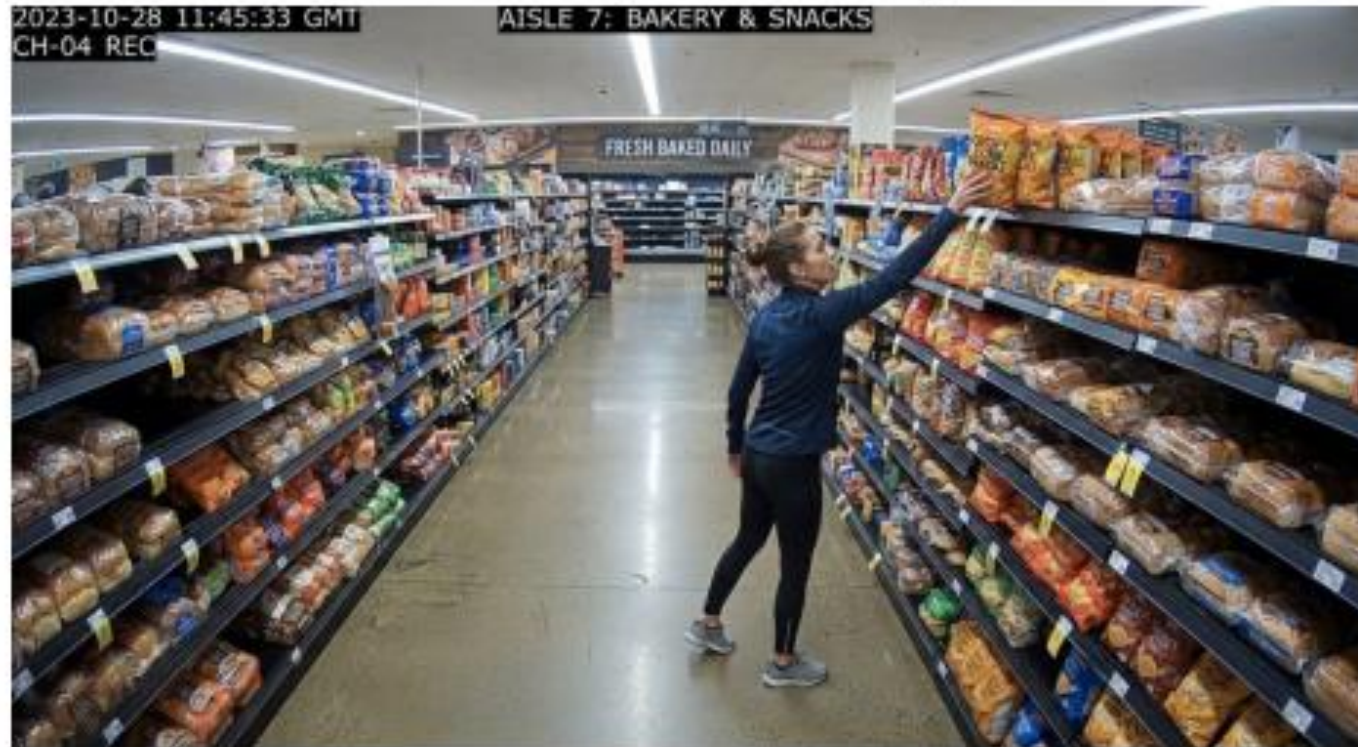
01



Input frame

CCTV frame or
generated image

1. Raw uploaded image



From raw frame to a 224×224 crop: an example

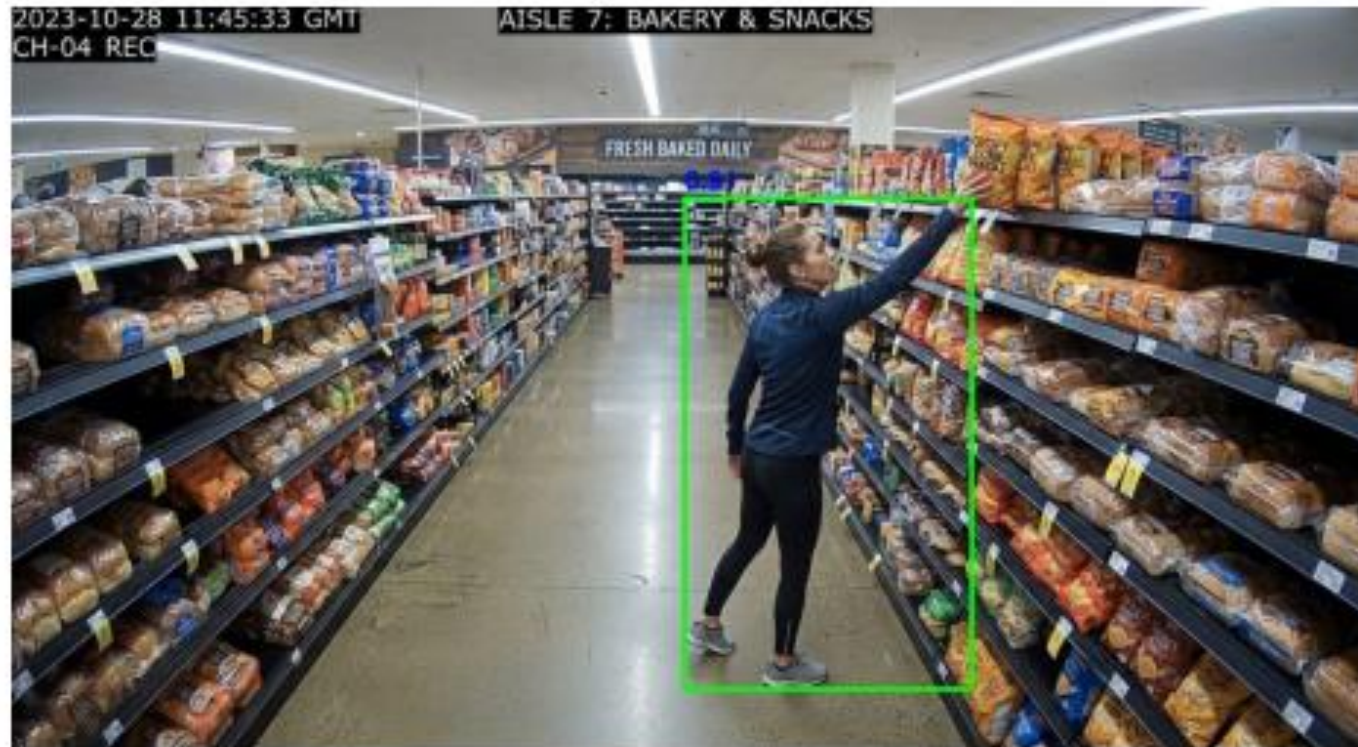
02



YOLO detect

person class,
conf ≥ 0.25

2. YOLO person detection



From raw frame to a 224×224 crop: an example

03



Expand box

+20–30% for
context

3. Expanded bounding box (orange) vs. original (green)



From raw frame to a 224×224 crop: an example

04



Crop person

highest-conf
box only

4. Cropped person region (385x647)



From raw frame to a 224×224 crop: an example

05



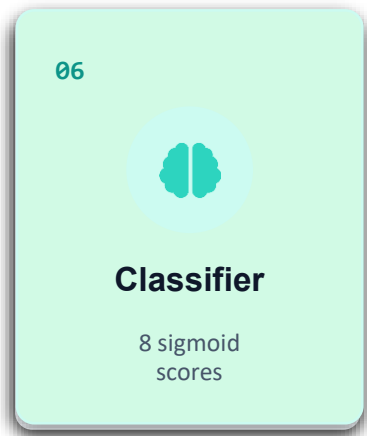
Letterbox

224×224, aspect
preserved

5. Letterboxed 224x224 RGB



From raw frame to a 224×224 crop: an example



Per-label results (sorted by probability):

[✓]	interacting_with_shelf	logit=+11.187	prob=1.0000
[]	no_visible_hand_interaction	logit=-9.238	prob=0.0001
[]	both_hands_not_visible	logit=-9.430	prob=0.0001
[]	hand_under_clothing	logit=-10.172	prob=0.0000
[]	hand_in_pocket	logit=-10.371	prob=0.0000
[]	hand_occluded_generic	logit=-10.377	prob=0.0000
[]	object_in_hand	logit=-10.542	prob=0.0000
[]	hand_in_bag	logit=-10.958	prob=0.0000

Final answer (probability ≥ 0.50):

- interacting_with_shelf (1.0000)

Augment first, then run the pipeline



The key design decision

Augmentations are applied to the raw images *before* YOLO detection — not after cropping.



320

original generated images



1,920

images after 5× augmentation



8 × 240

perfectly balanced classes

A split that can't leak

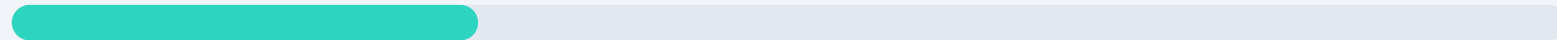
70 / 15 / 15 SPLIT — ON ORIGINALS

Train **1,344**



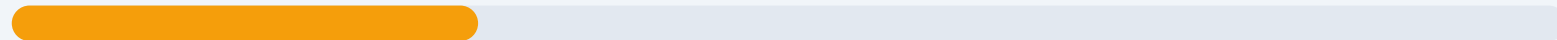
224 originals + 1,120 augmented

Val **48**



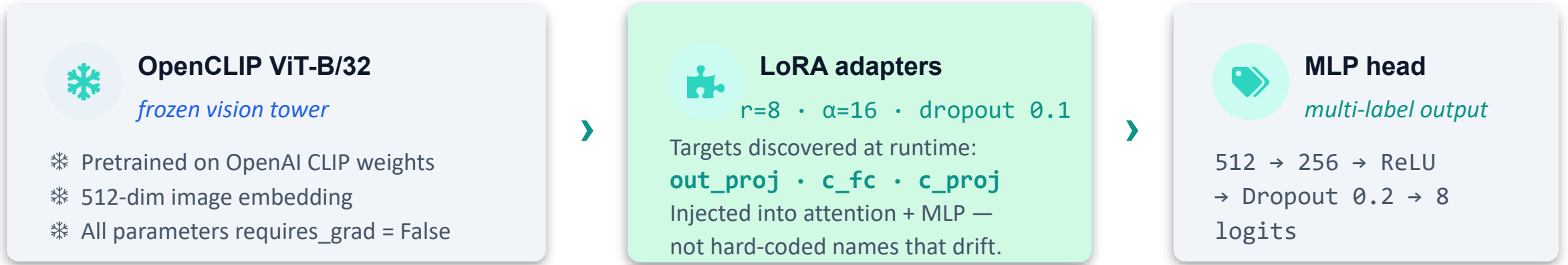
clean originals only

Test **48**



clean originals only

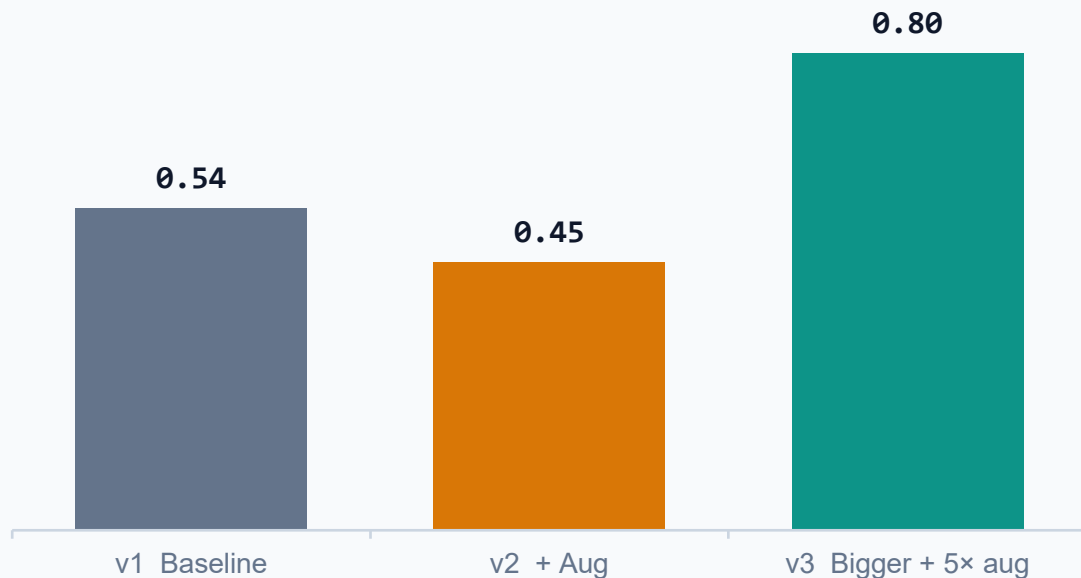
Frozen backbone, LoRA adapters, small head



Only the LoRA adapters and head learn; the CLIP tower stays frozen. The same recipe applies to every backbone next.

Three experiments to a working model

TEST MACRO-F1 BY ITERATION



v1

Baseline

F1 0.54

160 images, OpenCLIP + LoRA. Proved the pipeline end-to-end.

v2

Augmentation alone

F1 0.45

More augmented data, same small source set — generalization actually dipped.

v3

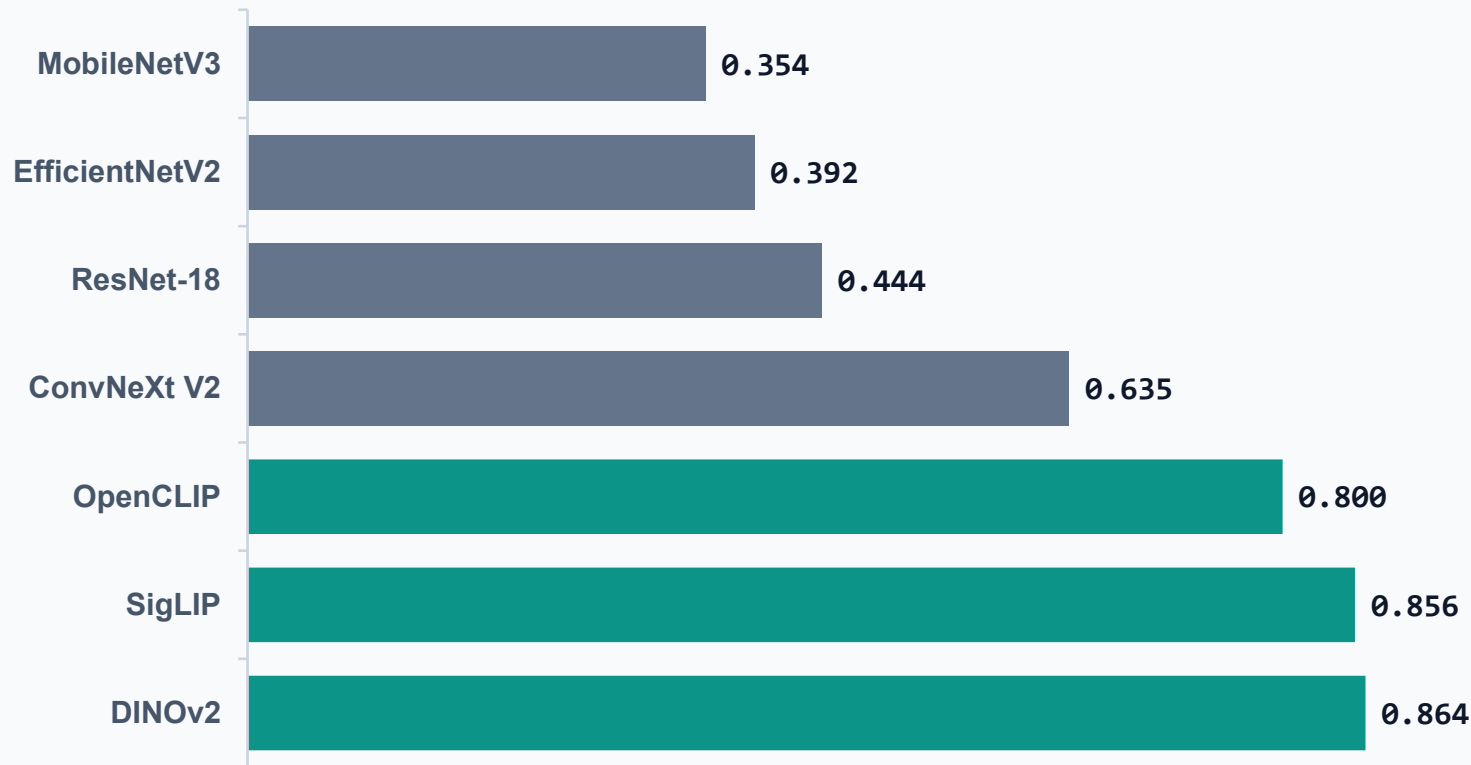
Scale + diversity

F1 0.80

Doubled the generated set and added 5x augmentation. Everything jumped.

Seven backbones, one identical protocol

Same data, same split, same head, loss, optimizer and epochs. Only the vision backbone changes. Bars are test-set macro-F1.



DINOv2 ViT-B/14

0.864

best macro-F1 · mAP 0.908

THE PATTERN

Foundation ViT + LoRA

sweeps the top three.

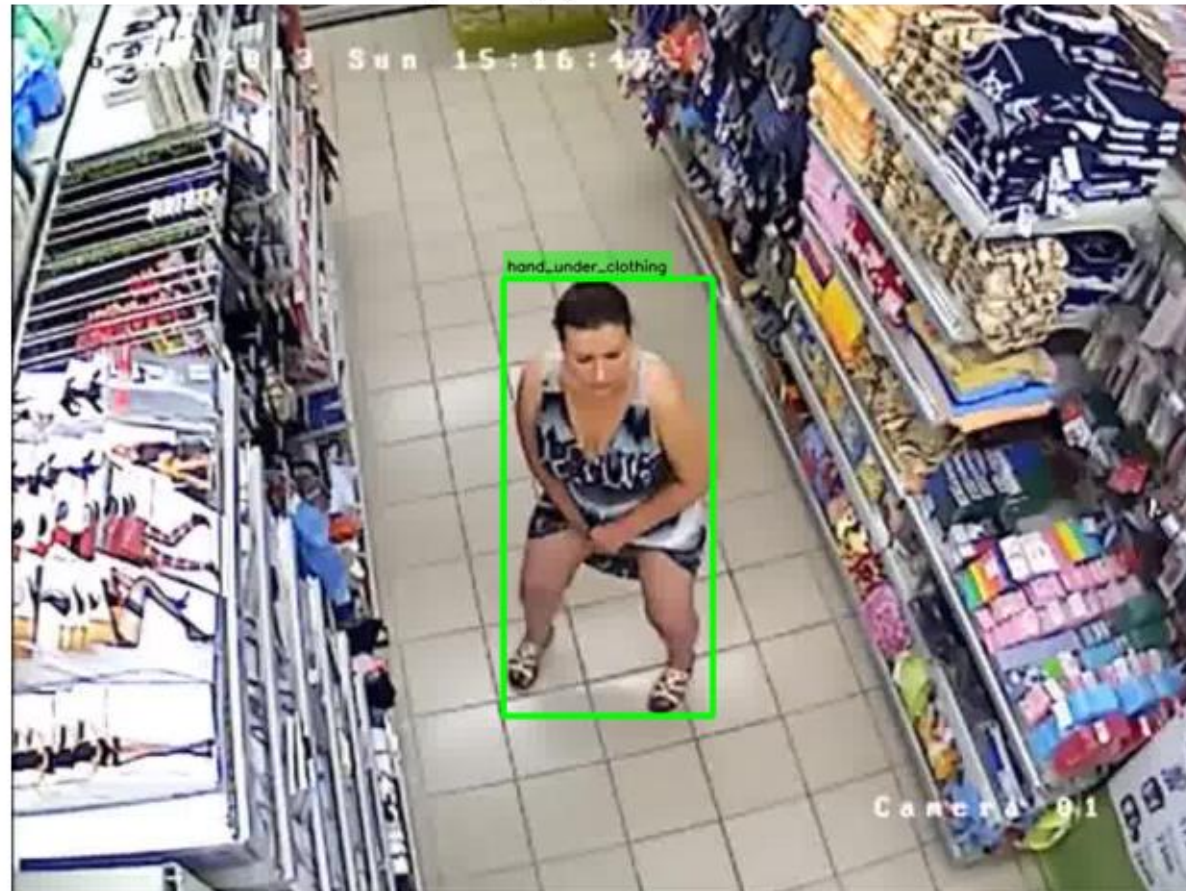
Frozen CNNs & from-scratch
trail well behind on so little
data.

Test-set metrics, every backbone

Backbone	Adaptation	F1	mAP	Subset acc	Trainable
DINOv2 ViT-B/14	Frozen + LoRA + head	0.864	0.908	0.833	1.58%
SigLIP ViT-B/16	Frozen + LoRA + head	0.856	0.882	0.750	1.42%
OpenCLIP ViT-B/32 <i>★ baseline</i>	Frozen + LoRA + head	0.800	0.857	0.750	0.67%
ConvNeXt V2-Tiny	Frozen + head	0.635	0.706	0.500	0.71%
ResNet-18	From scratch	0.444	0.436	0.375	100%
EfficientNetV2-S	Frozen + head	0.392	0.514	0.083	1.61%
MobileNetV3-Large	Frozen + head	0.354	0.414	0.125	7.70%

Evaluation on real images

Detected 1 person(s) with classification labels



[illegible]

What the project shows



Pretraining is the deciding factor



Diversity beats raw augmentation



Careful evaluation, honest numbers

Project Links



Scripts, Notebooks etc.

<https://github.com/OriMaimon2/SentinelFrame>



Generated Images and DINOv2 checkpoint

https://huggingface.co/datasets/OriMaimon/SentinelFrame_Generated_Images



Real datasets that been used

<https://universe.roboflow.com/retailvisiondata/person-shopping-in-supermarket>

https://universe.roboflow.com/amuletspace/cf_4cam_verano_con_pies

<https://universe.roboflow.com/school-hrk6p>

Frame-level today, behaviour-level tomorrow



Person tracking

link detections across frames



Temporal aggregation

fuse a track's frame states
over time



Risk scoring

turn patterns into a
calibrated signal



Real-time deployment

run the pipeline live on CCTV